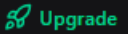


BANK DATABASE: SQL VIEWS

COMMON DATABASE SETUP

 OneCompiler 
 queries.sql × +


```
1 CREATE TABLE Customer (  
2     Customer_ID INT PRIMARY KEY,  
3     Customer_Name VARCHAR(100),  
4     Phone VARCHAR(15),  
5     Email VARCHAR(100),  
6     City VARCHAR(50)  
7 );  
8  
9 CREATE TABLE Account (  
10     Account_No INT PRIMARY KEY,  
11     Customer_ID INT,  
12     Account_Type VARCHAR(20),  
13     Balance DECIMAL(12,2),  
14     Branch VARCHAR(50),  
15     FOREIGN KEY (Customer_ID)  
16         REFERENCES Customer(Customer_ID)  
17 );  
18  
19 CREATE TABLE Bank_Transaction (  
20     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,  
21     Account_No INT,  
22     Transaction_Type VARCHAR(20),  
23     Amount DECIMAL(12,2),  
24     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,  
25     FOREIGN KEY (Account_No)  
26         REFERENCES Account(Account_No)  
27 );  
28  
29 CREATE TABLE Loan (  
30     Loan_ID INT PRIMARY KEY,  
31     Customer_ID INT,  
32     Loan_Type VARCHAR(30),  
33     Loan_Amount DECIMAL(12,2),  
34     Interest_Rate DECIMAL(5,2),  
35     FOREIGN KEY (Customer_ID)  
36         REFERENCES Customer(Customer_ID)  
37 );
```




```

9 CREATE TABLE Loan (
10     Loan_Type VARCHAR(50),
11     Loan_Amount DECIMAL(12,2),
12     Interest_Rate DECIMAL(5,2),
13     FOREIGN KEY (Customer_ID)
14     REFERENCES Customer(Customer_ID)
15 );
16
17
18
19 INSERT INTO Customer VALUES
20 (101,'K ANUSRI','9876543210','anusri@gmail.com','Hyderabad'),
21 (102,'K RITHWIK','9876543211','rithwik@gmail.com','Vijayawada'),
22 (103,'M SREEYA','9876543212','sreeya@gmail.com','Bangalore'),
23 (104,'K ADVAITH','9876543213','advaith@gmail.com','Chennai'),
24 (105,'E HEMANTH','9876543214','hemanth@gmail.com','Hyderabad');
25
26 INSERT INTO Account VALUES
27 (10001,101,'Savings',50000,'Hyderabad'),
28 (10002,102,'Savings',75000,'Vijayawada'),
29 (10003,103,'Current',120000,'Bangalore'),
30 (10004,104,'Savings',45000,'Chennai'),
31 (10005,105,'Current',90000,'Hyderabad');
32
33 INSERT INTO Bank_Transaction
34 (Account_No,Transaction_Type,Amount)
35 VALUES
36 (10001,'DEPOSIT',10000),
37 (10001,'WITHDRAW',5000),
38 (10002,'DEPOSIT',15000),
39 (10003,'WITHDRAW',20000),
40 (10004,'DEPOSIT',5000);
41
42 INSERT INTO Loan VALUES
43 (501,101,'Home Loan',5000000,7.5),
44 (502,102,'Education Loan',1000000,6.5),
45 (503,103,'Car Loan',800000,8.2),
46 (504,104,'Personal Loan',500000,10.5),
47 (505,105,'Home Loan',4000000,7.2);
48

```

Q1 — Customer Account View

```

66 CREATE VIEW Q1_Customer_Account_View AS
67 SELECT
68     C.Customer_ID,
69     C.Customer_Name,
70     A.Account_No,
71     A.Account_Type,
72     A.Balance
73 FROM Customer C
74 JOIN Account A
75 ON C.Customer_ID = A.Customer_ID;
76
77 SELECT * FROM Q1_Customer_Account_View;

```

Customer_ID	Customer_Name	Account_No	Account_Type	Balance
101	K ANUSRI	10001	Savings	50000.00
102	K RITHWIK	10002	Savings	75000.00
103	M SREEYA	10003	Current	120000.00
104	K ADVAITH	10004	Savings	45000.00
105	E HEMANTH	10005	Current	90000.00

Q2 — Savings Accounts With Balance Greater Than 50000

<pre>CREATE VIEW Q2_Savings_High_Balance AS SELECT * FROM Account WHERE Account_Type = 'Savings' AND Balance > 50000; SELECT * FROM Q2_Savings_High_Balance;</pre>		Output:				
Account_No	Customer_ID	Account_Type	Balance	Branch		
10002	102	Savings	75000.00	Vijayawada		

Q3 — Current Accounts From Hyderabad

<pre>CREATE VIEW Q3_Hyderabad_Current AS SELECT * FROM Account WHERE Account_Type = 'Current' AND Branch = 'Hyderabad'; SELECT * FROM Q3_Hyderabad_Current;</pre>		Output:				
Account_No	Customer_ID	Account_Type	Balance	Branch		
10005	105	Current	90000.00	Hyderabad		

Q4 — Top Balance Accounts

<pre>CREATE VIEW Q4_Top_Balance_Accounts AS SELECT Account_No, Customer_ID, Account_Type, Balance FROM Account ORDER BY Balance DESC; SELECT * FROM Q4_Top_Balance_Accounts;</pre>		Output:				
Account_No	Customer_ID	Account_Type	Balance			
10003	103	Current	120000.00			
10005	105	Current	90000.00			
10002	102	Savings	75000.00			
10001	101	Savings	50000.00			
10004	104	Savings	45000.00			

Q5 — Customers Who Have Taken Loans

<pre>CREATE VIEW Q5_Customer_Loan_View AS SELECT C.Customer_Name, C.City, L.Loan_Type, L.Loan_Amount FROM Customer C JOIN Loan L ON C.Customer_ID = L.Customer_ID; SELECT * FROM Q5_Customer_Loan_View;</pre>		Output:			
Customer_Name	City	Loan_Type	Loan_Amount		
K ANUSRI	Hyderabad	Home Loan	500000.00		
K RITIMIX	Vijayawada	Education Loan	1000000.00		
M SREEYA	Bangalore	Car Loan	800000.00		
K ADVAITH	Chennai	Personal Loan	500000.00		
E HEMANTH	Hyderabad	Home Loan	4000000.00		

Q6 — Customers With Loan Amount Greater Than 1,000,000

```
60 CREATE VIEW Q6_High_Loan_Customers AS
61 SELECT
62     C.Customer_ID,
63     C.Customer_Name,
64     L.Loan_ID,
65     L.Loan_Type,
66     L.Loan_Amount
67 FROM Customer C
68 JOIN Loan L
69 ON C.Customer_ID = L.Customer_ID
70 WHERE L.Loan_Amount > 1000000;
71
72 SELECT * FROM Q6_High_Loan_Customers;
```

Output:

Customer_ID	Customer_Name	Loan_ID	Loan_Type	Loan_Amount
101	K ANUSRI	501	Home Loan	5000000.00
105	E HEMANTH	505	Home Loan	4000000.00

Q7 — Total Balance for Every Branch

```
66 (504,104, 'Personal Loan',500000,10.5),
67 (505,105, 'Home Loan',4000000,7.2);
68
69 CREATE VIEW Q7_Branch_Total_Balance AS
70 SELECT
71     Branch,
72     SUM(Balance) AS Total_Balance
73 FROM Account
74 GROUP BY Branch;
75
76 SELECT * FROM Q7_Branch_Total_Balance;
```

Output:

Branch	Total_Balance
Hyderabad	140000.00
Vijayawada	75000.00
Bangalore	120000.00
Chennai	45000.00

Q8 — Average Balance for Each Account Type

```
CREATE VIEW Q8_Account_Type_Average AS
SELECT
    Account_Type,
    AVG(Balance) AS Average_Balance
FROM Account
GROUP BY Account_Type;

SELECT * FROM Q8_Account_Type_Average;
```

Output:

Account_Type	Average_Balance
Savings	56666.666667
Current	105000.000000

Q9 — Number of Customers in Each City

```
CREATE VIEW Q9_City_Customer_Count AS
SELECT
    City,
    COUNT(*) AS Number_of_Customers
FROM Customer
GROUP BY City;

SELECT * FROM Q9_City_Customer_Count;
```

Output:

City	Number_of_Customers
Hyderabad	2
Vijayawada	1
Bangalore	1
Chennai	1

Q10 — Branches With Total Balance Greater Than 200000

```
67 (505,105,'Home Loan',4000000,7.2);
68
69 CREATE VIEW Q10_Rich_Branches AS
70 SELECT
71     Branch,
72     SUM(Balance) AS Total_Balance
73 FROM Account
74 GROUP BY Branch
75 HAVING SUM(Balance) > 200000;
76
77 SELECT * FROM Q10_Rich_Branches;
```

Output:
Program did not output anything!

Q11 — All Deposit Transactions

```
CREATE VIEW Q11_Deposit_Transactions AS
SELECT
    Transaction_ID,
    Account_No,
    Amount,
    Transaction_Date
FROM Bank_Transaction
WHERE Transaction_Type = 'DEPOSIT';

SELECT * FROM Q11_Deposit_Transactions;
```

Output

Transaction_ID	Account_No	Amount	Transaction_Date
1	10001	10000.00	2025-08-26 08:16:11
3	10002	15000.00	2025-08-26 08:16:11
5	10004	5000.00	2025-08-26 08:16:11

Q12 — Withdrawal Transactions Greater Than 10000

```
69 CREATE VIEW Q12_High-Withdrawals AS
70 SELECT *
71 FROM Bank_Transaction
72 WHERE Transaction_Type = 'WITHDRAW'
73 AND Amount > 10000;
74
75 SELECT * FROM Q12_High-Withdrawals;
```

Output

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
4	10003	WITHDRAW	20000.00	2025-08-26 08:16:49

Q13 — Customer + Account + Transaction View

```
CREATE VIEW Q13_Customer_Transaction_View AS
SELECT
    C.Customer_Name,
    A.Account_No,
    A.Account_Type,
    T.Transaction_Type,
    T.Amount,
    T.Transaction_Date
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID
JOIN Bank_Transaction T
ON A.Account_No = T.Account_No;

SELECT * FROM Q13_Customer_Transaction_View;
```

Output

Customer_Name	Account_No	Account_Type	Transaction_Type	Amount	Transaction_Date
K. ANUSRI	10001	Savings	DEPOSIT	10000.00	2025-08-26 08:16:11
K. ANUSRI	10001	Savings	WITHDRAW	5000.00	2025-08-26 08:16:11
K. RITHMEK	10002	Savings	DEPOSIT	15000.00	2025-08-26 08:16:11
H. SREEYA	10003	Current	WITHDRAW	20000.00	2025-08-26 08:16:11
K. JENATH	10004	Savings	DEPOSIT	5000.00	2025-08-26 08:16:11

Q14 — Customer + Account + Loan View

Query		Output					
		Customer Name	Account No	Account Type	Balance	Loan Type	Loan Amount
<pre> 10 CREATE VIEW Q14_Customer_Loan_View AS 11 SELECT 12 C.Customer_Name, 13 A.Account_No, 14 A.Account_Type, 15 A.Balance, 16 L.Loan_Type, 17 L.Loan_Amount 18 FROM Customer C 19 JOIN Account A 20 ON C.Customer_ID = A.Customer_ID 21 JOIN Loan L 22 ON C.Customer_ID = L.Customer_ID; 23 SELECT * FROM Q14_Customer_Loan_View; </pre>		K. ARUSHI	10001	Savings	50000.00	Home Loan	5000000.00
		K. RITHWIK	10002	Savings	75000.00	Education Loan	1000000.00
		M. SREEYA	10003	Current	120000.00	Car Loan	800000.00
		K. ADARSH	10004	Savings	45000.00	Personal Loan	500000.00
		E. HERMITH	10005	Current	90000.00	Home Loan	4000000.00

Q15 — Annual Loan Interest View

Query		Output					
		Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Annual_Interest
<pre> 10 CREATE VIEW Q15_loan_Interest_View AS 11 SELECT 12 Loan_ID, 13 Customer_ID, 14 Loan_Type, 15 Loan_Amount, 16 Interest_Rate, 17 (Loan_Amount * Interest_Rate / 100) AS Annual_Interest 18 FROM Loan; 19 SELECT * FROM Q15_loan_Interest_View; </pre>		501	101	Home Loan	5000000.00	7.50	375000.0000000000
		502	102	Education Loan	1000000.00	6.50	65000.0000000000
		503	103	Car Loan	800000.00	8.20	65600.0000000000
		504	104	Personal Loan	500000.00	10.50	52500.0000000000
		505	105	Home Loan	4000000.00	7.20	288000.0000000000

Q16 — Loan Amount, Interest, Annual Interest and Total Amount

Query		Output						
		Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Annual_Interest	Total_Amount
<pre> 10 CREATE VIEW Q16_loan_Total_View AS 11 SELECT 12 Loan_ID, 13 Customer_ID, 14 Loan_Type, 15 Loan_Amount, 16 Interest_Rate, 17 (Loan_Amount * Interest_Rate / 100) AS Annual_Interest, 18 Loan_Amount + 19 (Loan_Amount * Interest_Rate / 100) AS Total_Amount 20 FROM Loan; 21 SELECT * FROM Q16_loan_Total_View; </pre>		501	101	Home Loan	5000000.00	7.50	375000.0000000000	5375000.0000000000
		502	102	Education Loan	1000000.00	6.50	65000.0000000000	1065000.0000000000
		503	103	Car Loan	800000.00	8.20	65600.0000000000	865600.0000000000
		504	104	Personal Loan	500000.00	10.50	52500.0000000000	552500.0000000000
		505	105	Home Loan	4000000.00	7.20	288000.0000000000	4288000.0000000000

Q17 — Branches With More Than Two Accounts

Query		Output	
<pre> 10 ((504,104,'Personal Loan',500000,10.5), 11 (505,105,'Home Loan',4000000,7.2)); 12 13 CREATE VIEW Q17_Multiple_Account_Branches AS 14 SELECT 15 Branch, 16 COUNT(*) AS Number_of_Accounts 17 FROM Account 18 GROUP BY Branch 19 HAVING COUNT(*) > 2; 20 21 SELECT * FROM Q17_Multiple_Account_Branches; </pre>		Program did not output anything!	

Q18 — Account Types With Total Balance Greater Than 200000

```
CREATE VIEW Q18_Account_Type_Balance AS
SELECT
    Account_Type,
    SUM(Balance) AS Total_Balance
FROM Account
GROUP BY Account_Type
HAVING SUM(Balance) > 200000;

SELECT * FROM Q18_Account_Type_Balance;
```

Output:

Account_Type	Total_Balance
Current	210000.00

Q19 — Customers With Balance Between 50000 and 100000

```
CREATE VIEW Q19_Balance_Range_View AS
SELECT
    C.Customer_ID,
    C.Customer_Name,
    A.Account_No,
    A.Account_Type,
    A.Balance
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID
WHERE A.Balance BETWEEN 50000 AND 100000;

SELECT * FROM Q19_Balance_Range_View;
```

Output:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance
101	K AMISRI	10001	Savings	50000.00
102	K RITHMIK	10002	Savings	75000.00
105	E HEMANTH	10005	Current	90000.00

Q20 — Highest Balance Account in Each Branch

```
CREATE VIEW Q20_Highest_Branch_Account AS
SELECT
    A.Branch,
    A.Account_No,
    C.Customer_Name,
    A.Account_Type,
    A.Balance
FROM Account A
JOIN Customer C
ON A.Customer_ID = C.Customer_ID
WHERE A.Balance = (
    SELECT MAX(A2.Balance)
    FROM Account A2
    WHERE A2.Branch = A.Branch
);

SELECT * FROM Q20_Highest_Branch_Account;
```

Output:

Branch	Account_No	Customer_Name	Account_Type	Balance
Vijayawada	10002	K RITHMIK	Savings	75000.00
Bangalore	10003	M SREEYA	Current	120000.00
Chennai	10004	K ADVAITH	Savings	45000.00
Hyderabad	10005	E HEMANTH	Current	90000.00